

Electrostatic Correlations and Temperature-Dependent Dielectric Constant Can Model LCST in Polyelectrolyte Complex Coacervation

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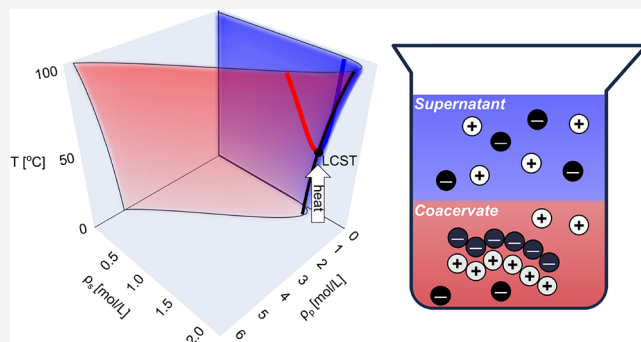


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ABSTRACT: The ability of polyelectrolytes to condense into a liquidlike, polyelectrolyte-rich phase out of a dilute supernatant phase through complex coacervation has led to fascinating phenomena, such as membraneless organelles and self-assembled capsules for drug delivery. Recent experiments have demonstrated that heating above a lower critical solution temperature (LCST) can drive complex coacervation. Here, we show that a coarse-grained model of electrostatic correlations is sufficient to model an LCST when accounting for the empirical decrease in the dielectric constant of the solvent upon heating. The predictions of the model agree qualitatively with experimental measurements of the compositions of the coexisting coacervate and supernatant phases. The model also achieves modest quantitative agreement with experiments, despite incorporating no other experimental parameters besides the dielectric constant and a fitted length scale. This agreement underscores the important role that can be played by electrostatic correlations in driving complex coacervation above an LCST.



1. INTRODUCTION

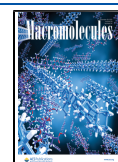
The electrostatic attraction between oppositely charged polyelectrolytes in solution can cause them to aggregate into a polyelectrolyte-rich coacervate phase and leave behind a polyelectrolyte-poor supernatant phase through a process known as “complex coacervation”. This phase separation underpins a variety of biological phenomena,^{1,2} from membraneless organelles^{3,4} to marine-life adhesion,^{5–10} and has been used to develop capsules for drug delivery^{11–13} and synthetic underwater adhesives.^{5,9,14–18} Voorn and Overbeek first identified the fundamental importance of electrostatic interactions in driving complex coacervation in the now well-known Voorn–Overbeek (VO) theory.¹⁹ Over 50 years later, Spruijt and co-workers demonstrated experimentally the predictions of VO theory regarding the effects of salt concentration and polymer chain length on the phase coexistence.²⁰ Experimental studies have since pushed the limits of VO theory by exploring the effects of hydrophobic interactions,²¹ pH,²² hydrogen bonding,^{22–24} and variations in monomer sequence.^{25–32} These studies have accordingly led to more sophisticated theories and simulations of complex coacervation, which have incorporated effects of charge regulation and ion-binding,^{21,33–41} excluded volume,^{42–46} dipole–dipole interactions,^{47,48} charge connectivity and electrostatic correlations,^{37,43–45,49–51} and charge sequence.^{30–32,52–54} These advances have also been documented in recent reviews on polyelectrolyte solutions.^{1,55–57}

The effect of temperature on the phase behavior of polyelectrolyte solutions, however, has received less attention. Much of the work on the effect of temperature has explored the apparent glass-transition temperature in solid polyelectrolyte precipitates.^{23,24,58,59} Here, we focus on the effect of temperature on the formation of liquidlike coacervates. While enthalpy can drive complex coacervation upon cooling below an upper critical solution temperature (UCST), such as in polyelectrolyte solutions with multivalent ions^{60,61} and some intrinsically disordered proteins,⁶² complex coacervation of strong polyelectrolytes is primarily driven by entropy^{63–65} when heated above a lower critical solution temperature (LCST), with an enthalpy of complexation that is either positive⁶⁶ or much smaller than the entropic gain in free energy,^{64,67} sometimes depending on the mixing order.⁶⁸ While some polyelectrolyte solutions exhibit an LCST because one of the components has an intrinsic LCST,^{11,16,17,69} recent studies have demonstrated an LCST in solutions of polyelectrolytes that do not exhibit an LCST on their own, such as solutions of hyaluronic acid and polypeptides,^{70,71} tau

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protein and RNA,^{53,72} hydrogels cross-linked by lanthanide coordination,⁷³ and proteins with specific sequences of amino acids.^{26,27} We focus on observations of an LCST in aqueous solutions of flexible, strong polyelectrolytes for a more straightforward comparison to the simplified systems amenable to theoretical models. Specifically, we focus on experiments performed with aqueous solutions of the polyanion poly(styrenesulfonate) (PSS), the polycation poly(diallyldimethylammonium) (PDADMA), and KBr salt. Wang and Schlenoff⁷⁴ first demonstrated phase separation upon heating such a solution using cloud-point measurements, which have since been supported by other measurement techniques.^{75–77} More recently, Ali et al.⁷⁵ reported the first measurements of the compositions of the coacervate and supernatant phases at different temperatures using UV–vis spectroscopy, with which they measured the concentration of PSS. These measurements were followed by the very recent work of Ma et al.,⁷⁷ in which they used ¹H NMR to measure the concentrations of both PSS and PDADMA in each of the coexisting phases.

Most studies attribute the entropy that drives complex coacervation to the release of either counterions^{22,30,39,45,46,63–67,78–83} or water molecules^{53,67,71,84} from the vicinity of lone polyelectrolyte chains upon complexation. The most appropriate way to account for the entropy of these species with theoretical models, however, remains unclear. While atomistic molecular dynamics simulations have elucidated the structure and interactions present during complex coacervation,⁶⁵ their computational resource requirements limit the detail with which they can predict phase behavior. To reduce the computational requirements, some studies model the solvent implicitly, but these may underpredict the measured entropy by a factor of up to 2 or 3.⁸³

While theoretical models^{35,36} and simulations^{45,46,65,79,80} have been developed to account for the entropy of counterions more explicitly, recent studies have shown that coarse-graining entropic degrees of freedom into other interactions can qualitatively model the phase behavior of complex coacervates upon heating above an LCST. For example, Adhikari et al.⁴⁸ qualitatively modeled the LCST measured by Ali et al.⁷⁵ by coarse-graining (1) hydrophobic solvent–polymer interactions into an entropic χ parameter and (2) electrostatic interactions into a dipole–dipole attraction term with an empirical model for the temperature dependence of the dielectric constant of water. Additionally, Lin et al.⁵³ and Park et al.⁷¹ modeled an experimentally observed LCST by coarse-graining the dehydration of the polyelectrolyte into a temperature-dependent excluded volume parameter in field theoretic simulations.

Here, we propose that electrostatic correlations are sufficient to drive phase separation of aqueous polyelectrolyte solutions upon heating above an LCST. We explore the phase behavior by modeling electrostatic correlations with a liquid state theory.^{44,85,86} We incorporate no other experimental parameters into our model besides the temperature dependence of the dielectric constant of water, which controls the strength of the electrostatic correlations, and a fitted length scale (the bead diameter). The temperature dependence of the dielectric constant is important because it leads to a Bjerrum length that increases, rather than decreases, with temperature, as noted by Ali et al.⁷⁵ and explored by Adhikari et al.⁴⁸ Indeed, by incorporating this temperature dependence, existing theories of the phase separation of polyelectrolyte solutions,^{31,32,47,87}

including the present liquid state theory,⁸⁸ predict an LCST rather than a UCST.

The rest of the work will proceed as follows. We first recapitulate the liquid state (LS) theory used in the work of Zhang et al.⁴⁴ in Section 2. Next, we present the complete binodal predicted by our LS theory in polyelectrolyte–salt–temperature space in Section 3.1. To demonstrate an LCST, we constrain the binodal surface to pass through the critical line at a single point by fixing the chemical potential of the salt ions to that of a theoretical saltwater reservoir in Section 3.2. To compare with experiments, however, we apply the constraints of a closed system with fixed overall concentrations of polyelectrolyte and salt using the method of Lagrange multipliers in Section 3.3. In this case, while the solution phase separates upon heating, it generally does not pass through an LCST but instead undergoes discontinuous phase separation. Finally, in Section 4, we compare the predictions of our LS theory with the experimental measurements of the composition of each phase reported by Ali et al.⁷⁵ and Ma et al.⁷⁷ Our model achieves qualitative agreement and a surprising, though modest, degree of quantitative agreement. Based on this agreement with experiments, we conclude that electrostatic correlations are sufficient to model an LCST in aqueous solutions of strong polyelectrolytes when accounting for the temperature dependence of the dielectric constant of water and, therefore, may play a dominant role in driving complex coacervation in such solutions.

2. THEORETICAL FORMULATION

2.1. Liquid-State Theory. In the present study, we model the phase behavior of polyelectrolyte solutions using a simple liquid-state (LS) theory^{85,86} based on the Helmholtz free energy density described in our previous work.⁴⁴ While the simpler model of electrostatic interactions proposed in VO theory is also sufficient to demonstrate an LCST when the temperature dependence of the dielectric constant of water is incorporated (see Figure S1), VO theory makes other predictions that are in strong disagreement with many experiments, such as steep, positively sloped tie lines in the polyelectrolyte–salt plane (while experiments typically measure zero or negative slope for strong polyelectrolytes^{46,89}) and salt thresholds beyond the limits of solubility.⁴⁵ Our LS theory, on the other hand, incorporates important physical interactions not included in VO theory, such as hard-sphere repulsion and connectivity of the polymer backbone. Although these are not essential for modeling phase separation above an LCST, they lead to better agreement with experiments.

In our model, we treat the polyelectrolyte (polycation and polyanion) and ionic (cation and anion) species as charged, hard spheres of uniform diameter σ , each with a charge of Z_i in units of the elementary charge. Because we are considering strong polyelectrolytes, for which the counterions completely dissociate, we specify that the charge per monomer $|Z_i| = 1$. We account for the connectivity of the hard spheres of the polyelectrolyte species according to the generalization of Wertheim's first-order thermodynamic perturbation theory (TPT1)^{90–92} to polyelectrolyte solutions, as presented by Jiang et al.^{85,86} We model the polycation and polyanion as chains of N_+ and N_- beads, respectively, and the ions as single beads. For simplicity, we will generally assume symmetry between polycation and polyanion (i.e., $N_+ = N_- = N$) and between cation and anion, except for the opposite signs of their charges. When comparing to experiments in Section 4, however, we will

relax this assumption and consider asymmetric polyelectrolyte chain lengths ($N_+ \neq N_-$). For further discussion of how asymmetric chain lengths modify our analysis, see Section S4 in the Supporting Information (SI).

We treat the solvent implicitly as a homogeneous dielectric medium. While a complete theory would explicitly account for the polarization of the solvent molecules,^{93–97} current explicit solvent models for complex coacervation⁴⁶ have not yet incorporated polarization effects. Given the approximation inherent in an implicit solvent model, we do not incorporate the salt concentration dependence observed in the bulk dielectric constant⁹⁸ because it would greatly complicate the analysis without achieving truly quantitative accuracy. For the sake of mathematical simplicity, we admit the limitations of our approximation and use the pure solvent dielectric constant. We do, however, incorporate the temperature dependence, which we found to be key to modeling an LCST. Following the example of Adhikari et al.,⁴⁸ we use Malmberg and Maryott's empirical model of the dielectric constant of water⁹⁹

$$\begin{aligned} \epsilon_r(T) = & 87.740 - 0.40008 T \\ & + 9.398 \times 10^{-4} T^2 \\ & - 1.410 \times 10^{-6} T^3 \end{aligned} \quad (1)$$

where T is the temperature in Celsius. Using this model for the temperature dependence of the dielectric constant results in a Bjerrum length that increases with temperature, whereas the Bjerrum length decreases with temperature if the dielectric constant is fixed (see Figure S3 in the SI).

In our LS theory, the Helmholtz free energy density is composed of an ideal (id) contribution and excess (ex) contributions

$$\beta f = \beta f^{\text{id}} + \beta f_{\text{hs}}^{\text{ex}} + \beta f_{\text{el}}^{\text{ex}} + \beta f_{\text{ch}}^{\text{ex}} \quad (2)$$

where β is the inverse of the absolute temperature scaled by the Boltzmann constant k_B . We briefly describe each contribution to the free energy below.

The ideal part of the free energy density describes the translational entropy of each component and is known exactly

$$\beta f^{\text{id}} = 2 \frac{\rho_p}{N} \left[\ln \left(\frac{\rho_p}{N} \Lambda_p^3 \right) - 1 \right] + 2 \rho_s [\ln(\rho_s \Lambda_s^3) - 1] \quad (3)$$

where ρ_i is the density and Λ_i is the thermal de Broglie wavelength of species i (p for polyelectrolyte and s for salt ion). Because Λ_i is independent of density, it is the same for each phase and, therefore, does not affect the phase behavior.

To model the nonideal, excess contributions to the free energy density, we first consider the system as an ensemble of *disconnected*, charged hard spheres. The contribution of hard-sphere repulsion to the excess free energy density can be accurately modeled using the Boublik–Mansoori–Carnahan–Starling–Leland (BMCSL) equation of state.^{100,101} In this study, we assume that the hard-sphere diameters are identical, $\sigma_p = \sigma_s = \sigma$, so the BMCSL theory simplifies to the classic Carnahan–Starling excess free energy density, $\beta f_{\text{hs}}^{\text{ex}} = 6\eta^2(4 - 3\eta)/[\pi\sigma^3(1 - \eta)^2]$,^{102,103} where $\eta \equiv (\pi/6)\sum_i \rho_i \sigma_i^3$ is the volume fraction of solute, with the sum taken over all ion and polyelectrolyte species i .

We model the contribution to the excess free energy density from electrostatic correlations using the Ornstein–Zernike equation with the mean spherical approximation (MSA) for

disconnected charged hard spheres.¹⁰⁴ For uniform hard spheres, this contribution to the excess free energy density simplifies to^{105,106}

$$\beta f_{\text{el}}^{\text{ex}} = - \frac{\Gamma^3(2/3 + \Gamma\sigma)}{\pi} \quad (4)$$

where the screening parameter Γ satisfies

$$\Gamma(1 + \Gamma\sigma) = \frac{\kappa}{2} \quad (5)$$

and where $\kappa \equiv \sqrt{4\pi l_B \sum_i \rho_i Z_i^2}$ is the inverse of the Debye screening length¹⁰⁷ and $l_B = \beta e^2/4\pi\epsilon_r\epsilon_0$ is the Bjerrum length, with e the elementary charge, ϵ_r the dielectric constant, and ϵ_0 the vacuum permittivity. While MSA was derived for dilute solutions, it has been shown to be quantitatively accurate for electrolyte solutions up to 2 M.¹⁰⁸

To account for the contribution of connectivity along the polymer backbone to the excess free energy density, we use the first-order thermodynamic perturbation theory (TPT1), initially proposed by Wertheim for an uncharged system^{90–92} and generalized to polyelectrolyte solutions by Jiang et al.^{85,86} While TPT1 is an approximate, perturbative approach to modeling the effects of connectivity, it has been shown to yield reasonable estimates of osmotic pressure in polyelectrolyte solutions.^{85,109} For the value of the monomer–monomer radial distribution function at hard-sphere contact, we use MSA combined with the exponential approximation,¹¹⁰ which results in a contribution to the excess free energy density of⁸⁶

$$\beta f_{\text{ch}}^{\text{ex}} = 2\rho_p \left(\frac{1}{N} - 1 \right) \ln y_{\text{pp}}(\sigma, \{n_\alpha\}) \quad (6)$$

where $y_{\text{pp}}(\sigma, \{n_\alpha\})$ is the cavity correlation function at contact estimated with the exponential approximation,¹¹⁰ as is commonly used for electrolyte solutions¹¹¹

$$\begin{aligned} y_{\text{pp}}(\sigma, \{n_\alpha\}) = & \left[\frac{1}{1 - n_3} + \frac{n_2\sigma}{4(1 - n_3)^2} \right] \exp \left(- \frac{\Gamma^2 a_p^2}{4\pi^2 \sigma l_B} \right) \\ & \times \exp \left(\frac{l_B Z_p^2}{\sigma} \right) \end{aligned} \quad (7)$$

where $n_2 \equiv \pi \sum_i \rho_i \sigma_i^2$, $n_3 \equiv \pi \sum_i \rho_i \sigma_i^3$, the volume fraction of solute, and $a_p \equiv \frac{2\pi l_B Z_p}{\Gamma(\Gamma\sigma + 1)}$ for uniform hard spheres. Although not essential for modeling an LCST, this contribution is included because excluding the effects of chain connectivity can lead to severe qualitative and quantitative disagreement with experiments, as is the case for VO theory.⁴⁵

2.2. Parameter Selection. To explore the predictions of our LS theory, we consider a symmetric solution of monodisperse, strong polyelectrolytes with a degree of polymerization $N = 100$, as in Adhikari et al.⁴⁸ Because the experiments of Ali et al.⁷⁵ used asymmetric polyelectrolytes, we relax the assumption of symmetric polyelectrolyte chain lengths when comparing our predictions to their measurements, as discussed in greater depth in Section 4. Having not found clear experimental measurements in the literature of the Kuhn length of the polyelectrolytes used in their experiments, we fit the bead size σ to their measurements. We found that $\sigma = 4.1$ Å minimized the absolute error taken across all data points and scaled by the experimental uncertainty of each data point. A qualitative demonstration of the sensitivity of the

phase behavior to these parameters is shown in Figure S2 in the SI.

2.3. Solving for Phase Coexistence. Although we consider a solution of five components (polycation, polyanion, cation, anion, and water), our assumptions of symmetric polyelectrolytes, symmetric salt ions, and an implicit solvent allow us to model the solution as a pseudo-two-component system of polyelectrolyte and salt. According to the Gibbs phase rule, a two-component system that separates into two phases has two degrees of freedom. Therefore, solving for the compositions of a pair of coexisting phases requires two constraints. In our LS theory, as in experiments, fixing the temperature is a convenient constraint to apply because it can be directly specified, in this case, through the Bjerrum length l_B . In the numerical implementation of our LS theory, the second constraint we apply is fixing the concentration of salt ions in the coacervate phase ρ_s^{co} . To compute a single pair of coexisting compositions, we apply these two constraints—fixed l_B and fixed ρ_s^{co} —to the system of three equations that follows from the conditions of equilibrium: (1) the equality of the chemical potential of polyelectrolyte, (2) the equality of the chemical potential of salt, and (3) the equality of the osmotic pressure in each phase. These three equations are sufficient to solve for the three remaining unspecified concentrations of (1) polyelectrolyte in the supernatant phase ρ_p^{sup} , (2) salt ions in the supernatant phase ρ_s^{sup} , and (3) polyelectrolyte in the coacervate phase ρ_p^{co} . To sweep out the full binodal surface, we vary the fixed value of ρ_s^{co} from zero to the critical concentration at each temperature. In this way, we sweep through all compositions along the binodal because ρ_s^{co} increases monotonically while approaching the critical point along the binodal.

Because experimental measurements of polyelectrolyte phase behavior are typically performed in beakers, we model them by considering a closed system with fixed overall composition. A closed system, however, does not in general pass through a critical point upon heating. Rather, it undergoes a discontinuous phase transition—unless the composition is specified to lie along the critical line a priori—at a temperature that we will call the “separation temperature”. Note that this theoretical separation temperature is lower than the experimentally measured “cloud-point temperature” because experimental observation of phase separation requires additional heating to speed up the kinetics.

An open system is thus more convenient than a closed system to study an LCST because it can exchange particles with an external reservoir to achieve the necessary critical composition. For this reason, we will first demonstrate an LCST in a polyelectrolyte solution that can exchange salt with an external saltwater reservoir. We apply this condition by specifying the chemical potential of the salt ions in the solution to be that of salt ions in a saltwater reservoir.

Later, to compare with experiments,^{75,77} we consider a closed system with fixed overall polyelectrolyte and salt concentrations. We apply these constraints to our model using the method of Lagrange multipliers. The constraints follow from the conservation of mass and the assumption of incompressibility

$$\phi \rho_p^{\text{sup}} + (1 - \phi) \rho_p^{\text{co}} = \rho_p \quad (8)$$

$$\phi \rho_s^{\text{sup}} + (1 - \phi) \rho_s^{\text{co}} = \rho_s \quad (9)$$

where ϕ is the volume fraction of the supernatant phase, ρ_p^{sup} is the concentration of polyelectrolyte in the supernatant phase, ρ_s^{sup} is the concentration of ions in the supernatant phase, and ρ_p^{co} and ρ_s^{co} give the corresponding concentrations in the coacervate phase. The overall free energy density of the two-phase solution can be expressed as the volumetric average, $\phi f^{\text{sup}} + (1 - \phi) f^{\text{co}}$, where f^{sup} is the free energy density of the supernatant phase and f^{co} is the free energy density of the coacervate phase. Thus, the resulting Lagrangian function is

$$L = \phi f^{\text{sup}} + (1 - \phi) f^{\text{co}} - \mu_p [\phi \rho_p^{\text{sup}} + (1 - \phi) \rho_p^{\text{co}}] - \mu_s [\phi \rho_s^{\text{sup}} + (1 - \phi) \rho_s^{\text{co}}] \quad (10)$$

with Lagrange multipliers μ_p and μ_s , which correspond to the chemical potentials of the polyelectrolyte and salt species.

With five independent variables, $\{\rho_p^{\text{sup}}, \rho_p^{\text{co}}, \rho_s^{\text{sup}}, \rho_s^{\text{co}}, \phi\}$, minimizing the Lagrangian function in eq 10 with respect to each variable yields five independent equations. Adding the two constraints results in a system of seven independent equations to solve for the equilibrium composition

$$\begin{cases} \frac{\partial f^{\text{sup}}}{\partial \rho_p^{\text{sup}}} = \mu_p \\ \frac{\partial f^{\text{co}}}{\partial \rho_p^{\text{co}}} = \mu_p \\ \frac{\partial f^{\text{sup}}}{\partial \rho_s^{\text{sup}}} = \mu_s \\ \frac{\partial f^{\text{co}}}{\partial \rho_s^{\text{co}}} = \mu_s \\ f^{\text{sup}} - \mu_p \rho_p^{\text{sup}} - \mu_s \rho_s^{\text{sup}} = f^{\text{co}} - \mu_p \rho_p^{\text{co}} - \mu_s \rho_s^{\text{co}} \\ \phi \rho_p^{\text{sup}} + (1 - \phi) \rho_p^{\text{co}} = \rho_p \\ \phi \rho_s^{\text{sup}} + (1 - \phi) \rho_s^{\text{co}} = \rho_s \end{cases}$$

If the overall polyelectrolyte and salt concentrations ρ_p and ρ_s are specified, as they are in the experiments by Ali et al.,⁷⁵ then this system of seven equations uniquely determines the seven unknowns: five independent variables and two Lagrange multipliers.

3. RESULTS

We first present the complete phase diagram plotted in polyelectrolyte–salt–temperature space. To demonstrate an LCST, we then consider an open system in chemical equilibrium with a saltwater reservoir. For comparison with experiments, we consider a closed system with fixed polyelectrolyte and salt concentrations.

3.1. Complete Phase Diagram. As discussed above, according to the Gibbs phase rule, our solution of symmetric polyelectrolyte and salt in implicit solvent has two degrees of freedom upon phase separation. The binodal is thus a two-dimensional manifold, which we call the binodal surface. We plot the binodal surface in polyelectrolyte–salt–temperature space in Figure 1, showing all thermodynamically stable compositions of supernatant (blue) and coacervate (red) phases possible for this system within the temperature range of liquid water (0–100 °C), with the critical line shown in black.

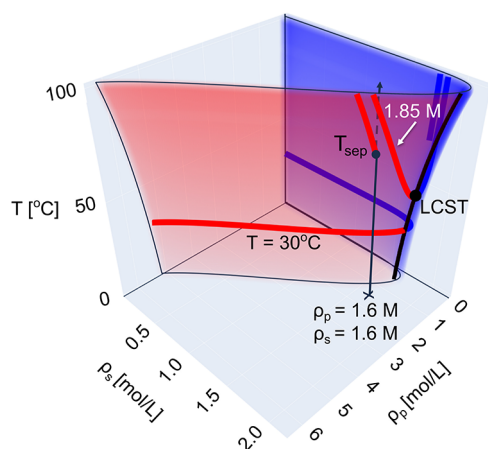


Figure 1. Binodal surface of a symmetric polyelectrolyte solution ($N = 100$) as a function of salt concentration ρ_s (mol/L), polyelectrolyte concentration ρ_p (mol/L), and temperature T (Celsius). The supernatant phase (blue surface) meets the coacervate phase (red surface) along the critical line (black line). An isotherm at 30 °C is highlighted with blue (supernatant) and red (coacervate) lines along a horizontal plane. A lower critical solution temperature (“LCST”, larger black circle) is achieved by placing the system in chemical equilibrium with a reservoir of saltwater at a concentration of 1.85 M, with the corresponding binodal highlighted with blue (supernatant) and red (coacervate) curves. When the overall concentrations of polyelectrolyte (1.6 M) and salt (1.6 M) are fixed (see the black “x”), heating the single-phase solution along the vertical black arrow leads to phase separation not at an LCST, but at a separation temperature T_{sep} (smaller black circle) upon intersecting with the binodal surface, at which the solution discontinuously separates into coacervate (red line) and supernatant (opposing blue line) phases.

Different constraints yield different slices of the binodal surface. For example, fixing the temperature to 30 °C and varying the salt concentration yields the curve parallel to the polyelectrolyte–salt plane highlighted in Figure 1. Fixing the overall concentrations of polyelectrolyte ρ_p and salt ρ_s , as was done in recent experiments,^{73,77} yields a curve with a gap rising upward along the temperature axis. Figure 1 shows such a path for $\rho_p = 1.6$ M and $\rho_s = 1.6$ M, where the vertical black arrow rising from the “x” on the polyelectrolyte–salt plane indicates the fixed overall composition as temperature increases. Due to the constraint on the overall composition, the system phase

separates through a discontinuous phase transition at a separation temperature T_{sep} . An open system, however, will phase separate through a continuous phase transition at a lower critical solution temperature (LCST). Figure 1 highlights the slice of the binodal for such a system placed in equilibrium with a saltwater reservoir having 1.85 M of salt. The phase behavior of open and closed systems will be discussed in greater detail in the following sections.

3.2. Open System. In an open system, the solution can exchange polyelectrolyte and salt with the environment, which may allow it to attain a critical composition. To demonstrate an LCST in a polyelectrolyte solution, we consider a semiopen system that allows for exchange of salt with a saltwater reservoir. Physically, this constraint could be realized by placing the solution in contact with a large tank of saltwater and separating it from the tank with a semipermeable membrane that permits salt ions—but not polyelectrolyte—to pass through. When the reservoir has an appropriate concentration of salt, our LS theory predicts an LCST. For example, a polyelectrolyte solution in chemical equilibrium with a reservoir of 1.85 M salt solution exhibits the LCST shown in Figure 1.

To demonstrate how the phase behavior changes with different concentrations of salt in the saltwater reservoir, Figure 2 shows slices along the binodal surface at different saltwater concentrations projected onto the three coordinate planes: (a) polyelectrolyte–temperature, (b) salt–temperature, and (c) polyelectrolyte–salt. At a low salt concentration (e.g., 1.60 M), the solution phase separates at all temperatures of liquid water (0–100 °C). The composition of the supernatant closely resembles that of the reservoir, having little polyelectrolyte (Figure 2a) and similar salt concentration (Figure 2b). In the range of critical salt concentrations (e.g., 1.75 and 1.90 M), the solution passes through an LCST upon heating. As the salt concentration in the reservoir increases, the width of the binodal projected onto the polyelectrolyte–temperature plane narrows (Figure 2a) and the LCST increases. As the temperature increases, the critical salt concentration and, to a lesser extent, the critical polyelectrolyte concentration increase as well. Temperature has a negligible effect on the slopes of the tie lines in the polyelectrolyte–salt plane, however, as one cannot distinguish between the tie lines

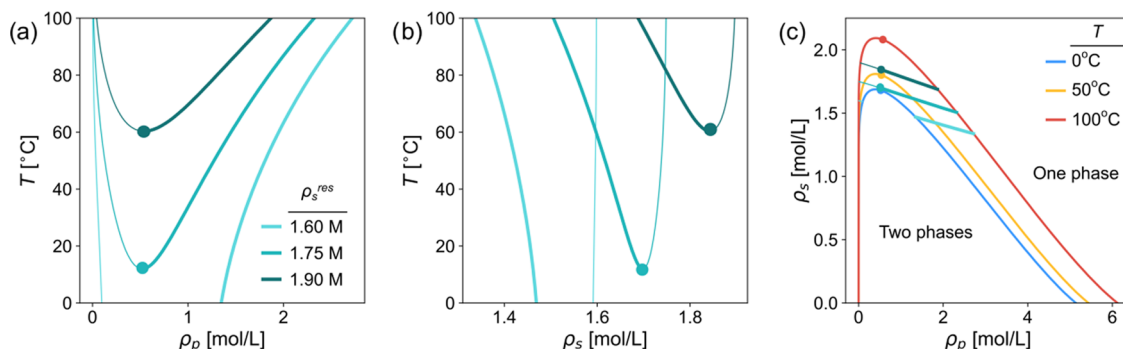


Figure 2. Projections of slices of the binodal surface for a semiopen system that can exchange salt with a saltwater reservoir at salt concentrations ρ_s^{res} of 1.60, 1.75, and 1.90 M, represented by increasingly dark shades of teal, with the coacervate branch represented by a thicker line than the supernatant branch. The binodals are projected onto three planes: (a) polyelectrolyte–temperature, (b) salt–temperature, and (c) polyelectrolyte–salt. At the two higher concentrations, 1.75 and 1.90 M, the supernatant and coacervate branches of the binodal meet at a lower critical solution temperature (LCST, filled circle). In the projection on the polyelectrolyte–salt plane, three isothermal cross sections of the binodal surface are shown at 0 °C (blue), 50 °C (orange), and 100 °C (red), along with their critical points (filled circles).

corresponding to different temperatures for a given reservoir salt concentration (Figure 2c).

3.3. Closed System. While an open system provides a clear demonstration of an LCST, most experiments are performed in a closed system with fixed overall concentrations of polyelectrolyte and salt. Here, we explore how the phase behavior changes under such constraints. We apply these constraints using the method of Lagrange multipliers, where the Lagrangian function is defined in eq 10. An example of a constrained binodal is shown in Figure 1, for which the overall polyelectrolyte and salt concentrations were both fixed to 1.60 M. Unlike the open system, which passes through a critical point, the closed system cannot achieve a critical composition because the overall composition is fixed to a noncritical value, and passes through a discontinuous phase transition at a separation temperature T_{sep} . This constraint may therefore lead to a gap between the supernatant and coacervate branches of the binodal, as seen in Figure 1 at the separation temperature T_{sep} .

To explore the phase behavior of the closed system more thoroughly, we plot the coexistence curves of three representative compositions in Figure 3. As in Figure 2, we project these curves onto the three coordinate planes: (a) polyelectrolyte–temperature, (b) salt–temperature, and (c) polyelectrolyte–salt, where the overall compositions are marked by glyphs superposed on isotherms from the binodal

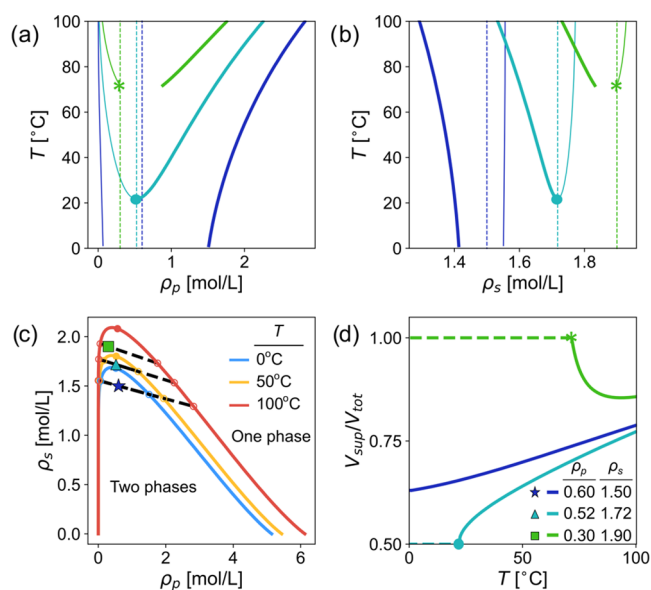


Figure 3. Projections of constrained binodals for closed systems with fixed overall compositions onto (a) the polyelectrolyte–temperature plane, (b) the salt–temperature plane, and (c) the polyelectrolyte–salt plane. The polyelectrolyte ρ_p and salt ρ_s concentrations (mol/L) corresponding to each glyph (blue star, teal triangle, and green square) are listed in (d). In (a) and (b), the coacervate branch is distinguished from the supernatant by a thicker line, and the overall composition is indicated by a thin vertical dashed line of the corresponding color. Panel (c) includes isotherms from the binodal surface at 0 °C (blue), 50 °C (orange), and 100 °C (red), with critical points marked by a filled circle. Panel (d) plots the volume fraction of the supernatant phase $V_{\text{sup}}/V_{\text{tot}}$ as a function of temperature and lists the compositions of each system. The horizontal dashed lines (green and teal) indicate a single phase. Throughout the figure, a continuous phase transition through a critical point is marked by a filled circle, whereas a discontinuous phase transition is marked by an asterisk.

surface in (c). Because the total volume is fixed in a closed system, we also plot the volume fraction of the supernatant phase as a function of temperature in (d).

The phase behavior of each of the three compositions plotted in Figure 3 can be understood based on the location of the composition relative to the binodal surface. First, the composition with the lowest salt concentration ($\rho_p = 0.60$ M, $\rho_s = 1.50$ M, blue star) resides within the two-phase region of the binodal at all liquid water temperatures (Figure 3c), so the system is always phase-separated into supernatant and coacervate phases. As temperature increases, the coacervate phase becomes richer in polyelectrolyte (Figure 3a) and poorer in salt (Figure 3b) and occupies a smaller volume fraction of the system (Figure 3d). Second, the composition with the highest salt concentration ($\rho_p = 0.30$ M, $\rho_s = 1.90$ M, green square) resides in the one-phase region at 0 and 50 °C but is in the two-phase region at 100 °C in Figure 3c. Because this composition does not lie on the critical line, the solution will discontinuously phase separate at the separation temperature marked by a green asterisk in Figure 3. The constraint on the overall composition also leads to a gap between the supernatant and coacervate branches of the binodal at the transition temperature (Figure 3a,b), as in the projected binodal slices shown in the study by Adhikari et al.⁴⁸ When cooling through this transition, the volume fraction of the supernatant initially decreases but, near the transition, the coacervate phase is depleted and the system becomes entirely supernatant, as seen by the rapid increase in supernatant volume fraction to 1 in Figure 3d. Finally, the composition with the intermediate salt concentration ($\rho_p = 0.52$ M, $\rho_s = 1.72$ M, teal triangle) resides at a critical point. In this case, the solution passes continuously through an LCST upon heating. Accordingly, there are no gaps in the binodal but, rather, the supernatant and coacervate branches meet at the critical point (teal circle in Figure 3a,b). Because the supernatant and coacervate phases are indistinguishable at the critical point, the system approaches a 50:50 mixture of supernatant and coacervate near the critical temperature before becoming a single, critical phase (Figure 3d).

Having shown representative examples of phase behavior in closed systems, we now examine the effects of systematically varying polyelectrolyte concentration and salt concentration on the phase diagram. Figure 4 compares the effects of varying the concentration of each of the components—(a) polyelectrolyte and (b) salt—within the polyelectrolyte–salt plane. In Figure 4, we see that increasing the salt concentration narrows the binodal along the polyelectrolyte axis. Additionally, the slopes of the tie lines become steeper at higher salt concentrations.

The effect of adding polyelectrolyte depends on the salt concentration. At the lowest salt concentrations, the tie lines are approximately horizontal (see Figure 4a), so adding polyelectrolyte simply shifts the composition to the right along the same tie line. At intermediate concentrations of salt, because the tie lines become negatively sloped, increasing the polyelectrolyte concentration raises the tie line along the salt axis (see Figure 4b), as with the addition of salt. Thus, adding polyelectrolyte in this regime likewise reduces the difference in polyelectrolyte concentration between supernatant and coacervate phases due to the narrowing of the binodal, but increases the difference in salt concentration due to the steepening of the tie-line slopes. At a high enough polyelectrolyte concentration, the system will become a single

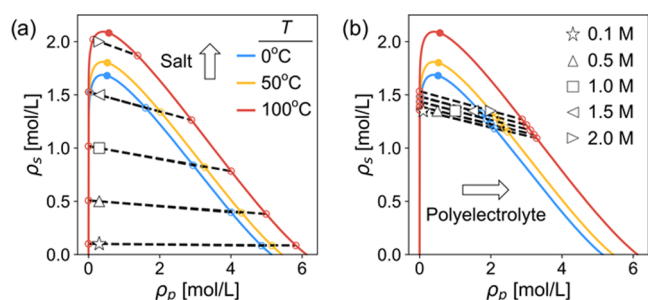


Figure 4. Different overall compositions (open symbols) of a closed system are plotted in the polyelectrolyte–salt plane with tie lines (dashed lines) indicating the coexisting compositions (open circles) of the supernatant (left branch) and coacervate (right branch) superposed on isotherms of the binodal surface. In (a), the overall polyelectrolyte concentration is fixed at $\rho_p = 0.3$ M and the overall salt concentration ρ_s is increased from 0.1 to 2.0 M (compositions listed in panel (b)). In (b), the overall salt concentration is fixed at $\rho_s = 1.35$ M and the overall polyelectrolyte concentration ρ_p is likewise increased from 0.1 to 2.0 M (compositions listed in top right). The legend in (b) only indicates the concentration of the varied component.

phase, either by passing through a critical point or depleting the supernatant.

4. DISCUSSION

Having presented the predictions of our LS theory, we now discuss them in the context of the experiments by Prabhu and co-workers^{75,77} as a test of how well a model of electrostatic correlations can capture the observed phase behavior. We begin by focusing on the effect of adding salt, as studied experimentally by Ali et al.⁷⁵ The polycation and polyanion used in the experiment were asymmetric (PDADMA: $M_n \approx 22$ kg/mol, $N_+ \approx 100$; KPSS: $M_n \approx 70$ kg/mol, $N_- \approx 400$ ¹¹²). We account for this asymmetry by relaxing our assumption of symmetric polyelectrolytes in our LS theory and selecting chain lengths similar to those measured experimentally ($N_+ = 100$ and $N_- = 400$). Additional details on how our computations changed to accommodate asymmetric chain lengths can be found in Section S4 in the SI.

Because the overall salt concentrations used in these experiments neared or exceeded the threshold salt concen-

tration predicted by our LS theory (≈ 2 M), an exact quantitative comparison was not fruitful. Rather, for the sake of illustration, we compare the experimental measurements with predictions by our LS theory made at salt concentrations that are 10% lower than the reported experimental values (the polyelectrolyte concentration remains unmodified at 0.3 M). These predictions are superposed on the measurements of polyanion (PSS) concentration reported by Ali et al. in Figure 5.

The phase behavior predicted by our LS theory generally agrees qualitatively with the experimental measurements of Ali et al.,⁷⁵ and the predicted polyanion concentrations are often within the uncertainty of the measured values. At the lowest salt concentration (Figure 5a), both experiments and theory find that the solution is phase-separated at all liquid water temperatures. While our LS theory overpredicts the polyanion concentration in the coacervate phase by almost double, its prediction for the supernatant phase is within experimental uncertainty. At the intermediate salt concentration (Figure 5b), although the experiments report a single phase below 5 °C where the theory predicts two phases, the predicted compositions are within experimental uncertainty for most temperatures. At the highest salt concentration, near the salt threshold (Figure 5c), both experiments and theory report a single phase below 10 °C. Although our LS theory significantly overestimates the polyanion concentration in the coacervate phase, its predicted concentrations in the supernatant phase are generally within experimental uncertainty. This level of agreement is surprising given that our LS theory includes no empirical parameters but the dielectric constant of the solvent and the fitted bead diameter, and it supports our hypothesis that electrostatic correlations can capture the observed phase behavior, at least qualitatively.

Adhikari et al.⁴⁸ also investigated the effect of overall salt concentration on phase coexistence, in their case using a mean-field model of dipole–dipole and solvent–polymer interactions. Their model also predicts that increasing the salt concentration raises the transition temperature and narrows the binodal along the polyelectrolyte axis, in agreement with our LS theory. However, the coacervate branches of the binodals presented in their work were all strictly convex, while our LS theory predicts the possibility of concavity at high temperature and low salt concentration (e.g., Figure 3a).

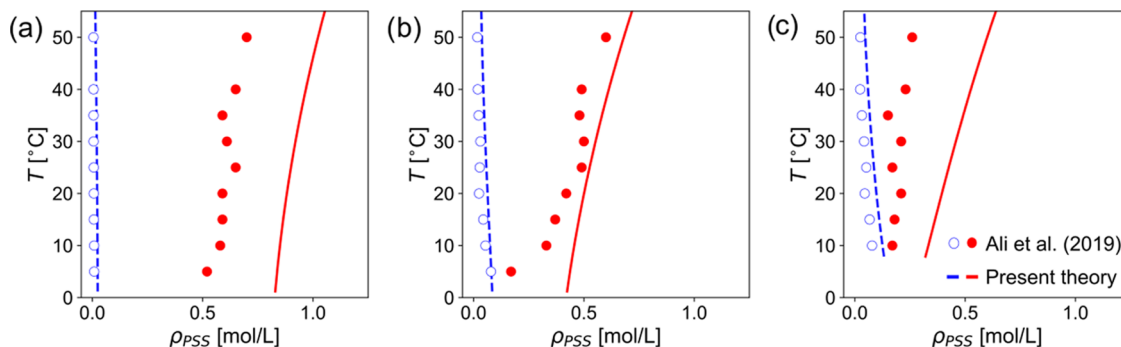


Figure 5. Comparison of experimental measurements by Ali et al.⁷⁵ with predictions by our LS theory of the polyanion (PSS) concentration in supernatant and coacervate phases at different temperatures. The overall polyanion concentration was 0.15 M (half of the overall polyelectrolyte concentration of 0.3 M). For the experimental measurements, the overall salt concentrations were (a) 1.75 M, (b) 2.00 M, and (c) 2.05 M. For our LS theory, the overall salt concentrations were reduced by 10% to (a) 1.58 M, (b) 1.80 M, and (c) 1.84 M. The values predicted by our LS theory are shown as lines (dashed blue for supernatant; red solid for coacervate) and the experimental measurements are shown as circles (open blue for supernatant; filled red for coacervate). Adapted with permission from ref 75. Copyright 2019, American Chemical Society.

Experimental measurements,^{75,77} however, have not yet definitively demonstrated a concave coacervate branch in the polyelectrolyte–temperature plane.

The effect of polyelectrolyte on the binodal has not been studied to the same extent. Ali et al.⁷⁵ reported that, upon increasing the polyelectrolyte concentration from 0.3 to 0.6 M at an intermediate salt concentration, the binodal narrowed along the polyelectrolyte axis (see Figure 3b in the main text and Figure S6 in the Supporting Information of their publication⁷⁵). This observation contradicts the predictions of our LS theory that show a widening of the binodal upon the addition of polyelectrolyte (see Figure 4b). Recent measurements from the same group,⁷⁷ however, reported a trend consistent with our LS theory.

The effect of polyelectrolyte concentration on the binodal is also related to the sign of the slope of the tie lines in the polyelectrolyte–salt plane, a question of interest to the field. In particular, the narrowing of the binodal in the polyelectrolyte–temperature plane upon increasing polyelectrolyte concentration, as predicted by our LS theory and measured by Ma et al.,⁷⁷ implies a negatively sloped tie line; the opposite trend, reported by Ali et al.,⁷⁵ implies a positively sloped tie line. This relationship is demonstrated graphically in Figure 6 and is further discussed by Ma et al.⁷⁷

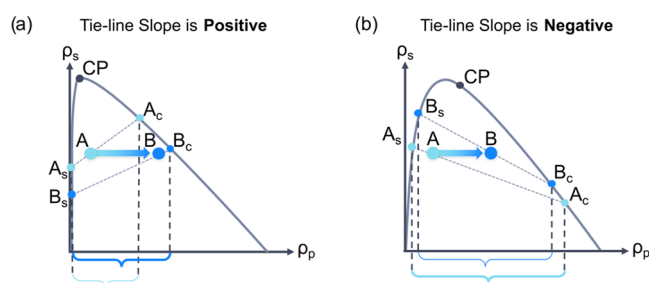


Figure 6. Schematic demonstrating that the addition of polyelectrolyte can (a) widen the binodal if the tie lines have a positive slope or (b) narrow the binodal if the tie lines have a negative slope, where the tie lines are shown by gray, sloped dashed lines. In each panel, the binodal (solid gray curve) is plotted as a function of polyelectrolyte composition ρ_p and salt composition ρ_s , with the critical point (CP) marked by a filled black circle. The addition of polyelectrolyte shifts the overall composition from point A (lighter blue) to point B (darker blue). This shift in composition changes the supernatant composition from A_s to B_s and the coacervate composition from A_c to B_c (small filled circles). The width of the binodal along the polyelectrolyte axis at each composition is spanned by horizontal brackets below the polyelectrolyte axis, with vertical dashed lines to guide the eye. The wider bracket has a thicker line for emphasis.

This relationship could offer an alternative experimental method for evaluating the sign of the slopes of the tie lines. Rather than measuring the salt concentration in both phases to determine the sign of the tie-line slope, an experimentalist could measure the polyelectrolyte concentration in just the coacervate phase at two different overall concentrations of polyelectrolyte. Given the limited number of experimentally measured tie-line slopes^{22,46,89} despite several theoretical predictions,^{33,42,45,51,113} an alternative measurement method could be valuable. Such measurements are increasingly important for our understanding of coacervation physics in light of recent findings that differences in the sign of the tie-line slope may indicate differences in the dominant interactions. Specifically, while negatively sloped tie lines

have generally been expected for solutions of strong polyelectrolytes in salt,¹¹³ positively sloped tie lines have now been shown at a low salt concentration (theoretically¹¹³ and experimentally⁸⁹), at a low polyelectrolyte charge fraction, and for more polar polyelectrolytes.⁸⁹

So far, our comparison between theory and experiment has only considered the concentration of the polyanion (PSS). Very recently, Ma et al. measured the concentrations of both polyelectrolytes (PSS and PDADMA) in phase-separated KBr solutions with ¹H NMR. They found that PSS preferentially partitions into the coacervate phase relative to PDADMA,⁷⁷ consistent with an earlier observation for centrifuged PSS–PDADMA complexes in NaCl solutions,¹¹⁴ and concluded that the cause is likely the greater solubility of PDADMA in water. This finding highlights the importance of incorporating species-specific χ parameters to capture differences in solubility for quantitative agreement, but this asymmetry does not affect the qualitative trends reported here.

Overall, the primary significance of the present work is our finding that electrostatic correlations and the temperature dependence of the dielectric constant of water are sufficient to model an LCST in a polyelectrolyte solution, as shown in Figure 1. While the LS theory used in this study also accounts for hard-sphere repulsion and chain connectivity, these are not required to drive phase separation by an LCST. In fact, VO theory, which accounts for neither, can also model an LCST (Figure S1 in the SI), although it incorrectly predicts steep, positively sloped tie lines.

Our results suggest two considerations for models of phase separation through an LCST in polyelectrolyte solutions. First, while many interactions may jointly contribute to phase separation through an LCST, electrostatic correlations are sufficient on their own. Determining their importance relative to other interactions in a given system, however, requires careful experimental investigation, like that in the work of Lin et al., which distinguished between hydrophobically and electrostatically driven phase separation in tau proteins.¹¹⁵ Second, accounting for the temperature dependence of the dielectric constant of water is essential to model an LCST with electrostatic correlations. The dielectric constant of water decreases so sharply with temperature (as a result of the entropic disruption of hydrogen bonds) that, in liquid water, the Bjerrum length increases with temperature, rather than decreasing (see Figure S3), indicating stronger electrostatic correlations upon heating. Consequently, a theoretical model of the phase behavior of polyelectrolyte solutions in water (or another solvent whose dielectric constant decreases sharply with temperature, such as many alcohols¹¹⁶) will underestimate the strength of electrostatic interactions at higher temperatures if it assumes a fixed dielectric constant. As mentioned in Section 1, the decision to incorporate the temperature dependence of the dielectric constant leads many theories,^{31,32,47,87} including our LS theory,⁸⁸ to predict an LCST instead of a UCST.

The importance of electrostatic correlations for phase behavior is further highlighted by the qualitative and modest quantitative agreement between the predictions of our LS theory and experimental measurements. Both theory and experiment show almost identical qualitative trends and, while the quantitative agreement is limited in scope and modest in degree, it is nonetheless surprising for three reasons. First, our LS theory does not incorporate any empirical parameters besides the dielectric constant of the pure solvent (water) and

a fitted bead diameter. Second, while MSA has been shown to give qualitatively accurate predictions of the electrostatic correlations in electrolyte solutions up to 2 M concentrations¹⁰⁸ and reasonable accuracy in polyelectrolyte solutions,^{50,51} Wertheim's first-order thermodynamic perturbation theory (TPT1) may not be accurate over as large a range of concentrations^{85,86,109} and requires further validation. While more rigorous theories of the coupling between chain connectivity and electrostatic correlations have been presented,^{50,51} TPT1 offers a simple and explainable approximation that is nevertheless sufficient to demonstrate the phase behavior in the present work. Third, experimental measurements have shown that entropy drives complex coacervation⁶⁴ and that this entropy primarily comes from the liberation of counterions^{45,80,82} or hydration shells^{71,84} upon complex coacervation. Our LS theory, however, does not account for either of these sources of entropy explicitly. Rather, the entropic degrees of freedom of the counterions are integrated into the coarse-grained expressions for the excess free energy and the solvent molecules are modeled implicitly through the dielectric constant. Nonetheless, our results show that this method of coarse-graining is sufficient to model an LCST. While the success of this coarse-grained model is encouraging, the extent to which counterion liberation and the entropy of solvent molecules must be modeled explicitly to achieve quantitative accuracy in predicting complex coacervation must still be addressed.

5. CONCLUSIONS

We used a liquid-state (LS) theory to demonstrate that electrostatic correlations are sufficient to drive phase separation of aqueous solutions of polyelectrolyte and salt upon heating above an LCST, finding qualitative and a surprising, though modest, degree of quantitative agreement with recent compositional measurements of aqueous solutions of poly(styrenesulfonate) (PSS), poly-(diallyldimethylammonium) (PDADMA), and KBr salt.^{75,77} Accounting for the temperature dependence of the dielectric constant of water was essential to model an LCST because it results in a Bjerrum length that increases, rather than decreases, with temperature, leading to stronger electrostatic correlations that drive phase separation at higher temperatures. We first demonstrated phase separation above an LCST by considering a semiopen system able to exchange salt with an external saltwater reservoir. To compare the predictions of our LS theory with experiments, we then considered a closed system with fixed overall concentrations of polyelectrolyte and salt and explored the effects of adding polyelectrolyte and salt on the phase behavior. Our predictions generally matched experimental results qualitatively, showing a narrowing of the binodal along the polyelectrolyte axis upon the addition of both salt and polyelectrolyte. This narrowing of the binodal upon addition of polyelectrolyte implies negative tie-line slopes in the polyelectrolyte–salt plane, opposite the preliminary finding of Ali et al.⁷⁵ but consistent with the more recent findings of Ma et al.⁷⁷ While a closer quantitative agreement between theory and experiment would likely require the inclusion of nonelectrostatic interactions, such as through a χ parameter,²¹ the surprising degree of quantitative agreement achieved using no other experimental parameters besides the dielectric constant and a fitted bead diameter underscores the important role that electrostatic correlation, as modified by the dielectric constant, can play in the phase behavior of

polyelectrolyte solutions. We hope that the present work encourages researchers that study the phase behavior of polyelectrolyte solutions to account for the temperature dependence of the dielectric constant in their models and consider the relative importance of electrostatic correlations in driving phase separation.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.macromol.1c02000>.

Computing phase compositions along the binodal (S1); lower critical solution temperature: Voorn–Overbeek theory (S2); effects of chain length, charge fraction, and bead size (S3); generalized phase diagram for asymmetric chain lengths (S4); Bjerrum length vs temperature: with and without temperature dependence of the dielectric constant of water (S5); and upper critical solution temperature (S6) (PDF)

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Notes

The authors declare no competing financial interest.

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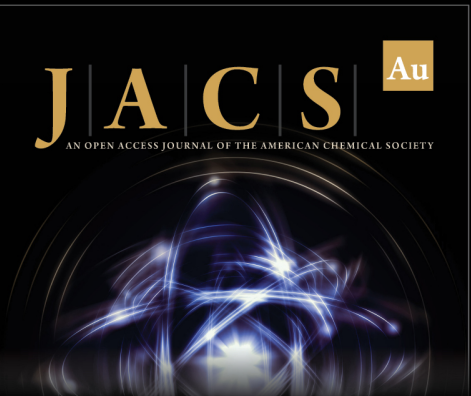
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